RAG Assignment: Baseline Retrieval vs. Cross-Encoder Re-Ranking

Document: ICC Men's Test Match Playing Conditions

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Code: https://github.com/darmojud/LMA/tree/RAG_Assignment

Setup

Both pipelines use the same corpus and the same vector index, so the only difference between them is the re-ranking step.

- **Corpus:** one PDF, 103 pages, 45,594 words, split into 326 overlapping chunks (180 words, 40-word overlap).
- **Pipeline 1 (baseline):** `all-MiniLM-L6-v2` (384-d) with a FAISS inner-product index on L2-normalised vectors (cosine). The top 10 chunks are retrieved and the top 3 go to the LLM, which gives Answer 1.
- **Pipeline 2 (re-ranking):** the same top 10, re-scored by `cross-encoder/ms-marco-MiniLM-L-6-v2` and re-sorted. The new top 3 go to the LLM with an identical prompt, which gives Answer 2.
- **LLM:** `llama3.1` through Ollama, temperature 0. All timings come from an Apple-silicon Mac (MPS).

1 Architecture comparison: Bi-Encoder vs. Cross-Encoder

	Bi-Encoder	Cross-Encoder
Input	Query and chunk are encoded *separately* by the same transformer.	Query and chunk are concatenated into one input: `[CLS] query [SEP] chunk [SEP]`.
Interaction	None until the end: cosine / dot product of two fixed vectors.	Joint self-attention: every query token attends to every chunk token.
Output	One vector per text.	One relevance score per (query, chunk) pair.
Pre-computable	Yes. Chunk vectors are indexed once, offline.	No. The score depends on the query, so each pair needs a forward pass.
Cost per query	One query encoding plus a vector search.	N forward passes for N candidates.
Role	High-recall first stage.	High-precision re-ranker.

Table 1: Core differences between the two encoders.

A bi-encoder squeezes a whole 180-word chunk into one vector. A chunk that only shares a topic or some vocabulary with the query can therefore outrank the chunk that holds the rule, and short decisive words ("cannot", "unless", "except") can carry little weight in the vector. A cross-encoder reads the query and the chunk together, token against token, so it is much better at telling which chunk actually answers the question. The cost is that nothing can be computed ahead of time.

2 Qualitative and rank-shift analysis

2.1 Query selection

I started with 26 candidate questions spread over the four failure categories. In 19 of them, re-ranking moved at least one chunk from baseline ranks 4–10 into the top 3. From these I kept five queries where Answer 1 was wrong or incomplete and Answer 2 fixed it (one per category, with negation used twice), and added one query that shows where re-ranking stops helping. I judged each answer by reading it against the retrieved clauses, and also ran a simple expected-words check, shown in the last column of Table 3.

2.2 Rank changes

Category	Query	Promoted into top 3 (baseline → re-ranked)	Dropped from top 3
Keyword trap	How much extra time can be used to complete the minimum overs?	Chunk 68 (overtime rule, cl. 12.7.1.1, p. 23): #10→#2	Chunk 83 (time-lost rules, cl. 12.11, p. 27): #1→#7
Exception	Can a team nominate a substitute fielder after the toss?	Chunk 3 (cl. 1.2.2, p. 4): #4→#1; chunk 9†: #5→#2	Chunk 264: #2→#4; chunk 147: #3→#6
Negation	Can a declaration or follow-on decision be withdrawn after it has been notified?	Chunk 91 (cl. 15.3, p. 30): #8→#2; chunk 297†: #7→#3	Chunk 287: #3→#6; chunk 20: #2→#8
Negation (2)	Are unbowled overs carried forward to the next day?	Chunk 68 (cl. 12.7.1.1, p. 23): #8→#3; chunk 66†: #5→#2	Chunk 73: #2→#4; chunk 67: #3→#6
Buried at 4–10	What is the penalty for running short deliberately?	Chunk 106 (cl. 18.5.2, p. 33): #4→#1	Chunk 103: #3→#4
Limitation	Can a captain declare in the final innings of the match?	Chunk 56†: #9→#2; chunk 90†: #7→#3. The answer chunk 91 was already #1.	Chunk 88: #2→#5; chunk 9: #3→#10

Table 2: Rank shifts of the top-10 candidates. † Promoted into the top 3 but does not contain the answer.

2.3 Answer 1 vs. Answer 2

Query	Answer 1 (baseline top 3)	Answer 2 (re-ranked top 3)	Check A1 / A2
Extra time	**Omitted the clause.** Says there is "no specific information" on extra time and cites hours-of-play and last-hour text.	Correct: 30 minutes of permitted overtime beyond the scheduled or rescheduled close.	No / Yes
Substitute after toss	**Omitted the exception.** A flat "No" based on cl. 24.1.1; does not mention cl. 1.2.2.	Finds the exception (the Match Referee may allow later additions in exceptional circumstances), but still concludes "not clear". Partial fix.	No / Yes
Declaration withdrawn	**False "not in context".** Says the document has nothing on declarations or follow-on decisions.	Correct: "No"; once notified, the decision cannot be changed (cl. 15.3).	No / Yes
Unbowled overs	**Hedged and misread.** Gives no definitive answer and reads cl. 12.7.8 as implying overs carry over.	Correct: "No"; unbowled overs are not made up on any subsequent day.	No / Yes
Short-run penalty	**Wrong.** Says no penalty is stated.	Correct: all runs disallowed, five penalty runs to the fielding side, plus the other consequences in cl. 18.5.2.	No / Yes
Declare in final innings	Says the context lacks the answer, although cl. 15.1 excludes the final innings from the right to declare.	Same failure: notes that an exception exists but says it is not specified.	No / No

Table 3: Comparison of the two answers. The check column is a substring test on a key phrase of the correct answer.

What the comparison shows.

- In all five cases the baseline did not make up a rule. Its mistakes were omissions: a missing exception or clause, or a false "the document does not say", plus one misreading (unbowled overs). In four of the cases (substitute fielder, declaration, unbowled overs, short-run penalty) the chunk with the answer sat at baseline ranks 4–8, and the cross-encoder moved it into the top 3. The keyword-trap case works differently, as explained next.
- **Keyword trap.** The bi-encoder ranked chunk 83 (the time-lost rules) first. The cross-encoder dropped it to #7 and promoted chunk 68, which uses the actual term ("permitted overtime"). The 30-minute figure itself is in chunk 67, which was already at baseline rank 3. For that reason the notebook's automatic verdict printed "No difference": a chunk containing "permitted overtime" was already in the baseline top 3. Answer 1 still said there was no information, so the gain here comes from removing the top-ranked distractor, not from retrieving a chunk that was missing.
- **Negation and exception.** Answer 1 for the declaration query was built from chunks about withdrawing an appeal, umpire signals and review lights. These are close in topic but are not the rule. Re-ranking brought in the clause that states the prohibition. For the substitute-fielder query, Answer 1 gave a flat "No", while Answer 2 found the exception.
- **Limitation.** For "declare in the final innings" the right chunk was already #1 in both pipelines, yet both answers failed. Re-ranking changes what the LLM sees, but it cannot fix how the LLM reads "with the exception of the final innings".

- **Not every promotion matters.** Among the 26 candidates, some queries changed rank without any real change in the answer (the bat-handle and new-ball questions, for example, gave 98% and 100% answer similarity). The benefit shows up when the answer-bearing chunk starts at ranks 4–10.

Caveats. Six queries and a single 8B-class LLM make this a small sample. It illustrates the effect but does not measure it statistically. The cross-encoder also only re-orders the same ten candidates, so a clause that falls outside the top 10 cannot be recovered.

3 Latency profiling

Stage	Mean (ms)	Median (ms)	Std (ms)	Min (ms)	Max (ms)
Pipeline 1: bi-encoder search	5.91	5.76	0.67	5.23	8.52
Pipeline 2 extra: cross-encoder on 10 chunks	51.54	50.52	5.35	42.02	60.52
Pipeline 2 total (search + re-rank)	57.46	56.01	5.50	47.55	67.17

Table 4: Retrieval-stage time: 6 queries × 5 repeats, warm-up run excluded.

Query	P1 retr. (ms)	P2 retr. (ms)	Re-rank (ms)	P1 LLM (s)	P2 LLM (s)	P1 total (s)	P2 total (s)
Keyword trap	96.95	150.62	53.67	5.13	1.20	5.22	1.35
Exception	53.94	117.27	63.32	7.39	9.31	7.44	9.42
Negation	68.84	128.71	59.87	2.91	3.04	2.98	3.17
Negation (2)	47.20	94.41	47.22	7.79	5.60	7.84	5.70
Buried at 4–10	62.66	122.82	60.16	4.54	8.32	4.60	8.44
Limitation	53.14	106.51	53.37	2.97	3.79	3.02	3.89
Average	63.79	120.06	56.27	5.12	5.21	5.19	5.33

Table 5: End-to-end time per query (single run each). P1 = baseline, P2 = with re-ranking; "Re-rank" is the extra time added by the cross-encoder.

- Re-ranking adds about 51.5 ms per query, or roughly 5.2 ms per (query, chunk) pair. That makes the retrieval stage about 9.7× slower than bi-encoder search alone (57.5 ms vs. 5.9 ms).
- End to end, the cross-encoder is about 56 ms of a 5.3 s Pipeline 2 query, roughly 1.1%. LLM generation takes nearly all the time in both pipelines (5.1 s vs. 5.2 s on average).
- The per-query differences in LLM time seem to follow the length of the generated answer, not the retriever. The one-line Answer 2 for the keyword-trap query, for example, took only 1.2 s. These differences should not be credited to re-ranking.
- Table 5 is made of single runs mixed in with LLM calls, so its retrieval columns are noisier and higher than those in Table 4. Table 4 is the better estimate of what each stage costs.

4 Scaling: why not run the Cross-Encoder over the whole collection?

A cross-encoder scores each (query, chunk) pair jointly, so nothing about a chunk can be computed ahead of time or indexed. Every query needs one transformer forward pass per chunk, and the cost grows linearly with the collection size N. At the measured 5.2 ms per pair, scoring all 326 chunks of this document would take about 1.7 s per query, compared with about 6 ms for bi-encoder search. For a million chunks the same linear extrapolation gives roughly 1.4 hours per query. Batching and faster hardware change the constant but not the linear growth.

A bi-encoder embeds every chunk once, offline. At query time it only encodes the query and compares vectors, and approximate-nearest-neighbour indexes such as HNSW or IVF make that search sub-linear in N. (The exact flat FAISS index used here is still linear in N, but each step is just a dot product, far cheaper than a transformer pass.) This is why the usual answer is the two-stage design used here. A cheap, high-recall retriever narrows the collection to a few candidates, and the slower, more precise cross-encoder re-ranks only those. The trade-off is that the re-ranker can only re-order what the first stage hands it, so the candidate set (the top 10 here) has to be large enough to contain the answer-bearing chunk.

Conclusion

On this document, re-ranking the same ten candidates moved the answer-bearing chunk from baseline ranks 4–8 into the top 3 in four of the six stress queries. In a fifth (the keyword trap) it replaced a top-ranked distractor. In all five, the final answer went from an omission or a false "not in context" to a correct one, though only partially for one of them. The extra cost is about 52 ms per query, around 1% of end-to-end time. Re-ranking does not help when the LLM misreads a clause that was retrieved correctly, or when the right chunk is outside the top 10.